

Project Proposal

Production Label Printing System (PLP) with QR Code — Phase 1

100% CUSTOMIZED SOFTWARE DEVELOPMENT OUTSOURCING

Poly Packaging Services | Prepared: May 2026 | Version 1.1

1. Project Overview

Poly Packaging Services requires a customized software solution for a Production Label Printing System (PLP) with QR Code functionality, purpose-built to align with their exact production workflow across five key sections: Mixing, Extruder, Printing, Gusset, and Cutting.

Unlike off-the-shelf software, this fully tailored system is designed around how Poly Packaging actually operates—not the other way around. This approach eliminates unnecessary features, removes the need for workarounds, and avoids ongoing licence fees to third-party vendors.

Advantages of 100% Custom Development

- Built exactly to PPS workflow — zero compromise on process fit
- Full control over future enhancements and phases
- Data stays within a private, dedicated cloud environment
- Faster adoption — staff learn a system built around their own terminology and process

The system will eliminate manual label generation, improve production traceability end-to-end, and provide real-time dashboards and reports for Directors and Managers — accessible from any device, including mobile, on the factory floor.

2. Monthly Cost

US\$60	Monthly "Low-Cost" Cloud Server (15 Users, 8 Concurrent Users) Initial 15GB for Database + Application (Excluding API Ports Connectivity to 3rd Party Other Software Systems used by the Client) **Equivalent LKR 800/- to 1000/- per day Cloud Server Space (Physical Server Cost is Rs. 2,500/- per day for min. 5 years life- time)
	Additional 5 GB Cloud Server Storage, Auto Backup, and RAID/Mirroring Service Features
	Monthly Software Support (bug fixes, security & framework updates, online support)
US\$100	OPTIONAL/RECOMMENDED CLOUD & SQL DB : US\$?? MICROSOFT AZURE - Monthly Cloud Server Charges (15 Users, 8 Concurrent Users) Initially 15GB for Database + Application (Excluding API Ports Connectivity to 3rd Party Other Software Systems used by the Client) **Equivalent LKR 2,000/- per day Server Space (Physical Server Cost is Rs. 2,500/- per month for min. 5 years life- time)

3. Phase 1 Scope

SOFTWARE DEVELOPMENT COST <i>(Excluding Requirements Analysis for Software Information Architecture and Software QA Testing before releasing to Users)</i>	100% CUSTOMIZED SOFTWARE FEATURE (As per PPS/PPE) <i>(Software Implementation, Limited TOT Training, Technical Support will be provided online. Customer can request for physical visit to the Factory Site and there will be an additional Fee for Professional/Engineer Hourly Fee and Transport per Linear Km)</i>
55,000	Digital Platform Multi-Tenants and Comp , APIs, Non-Functional Development, 2 Multi-Companies Database Setup, Software Initial Testing and Live Server Environment Setup, and Train-the-Trainer (3 Key/IT Users) Cost.
55,000	01. PHASE-1: User Management Module with Audit Trail Logs (Access to only 5 Directors/ IT level Users)(Multi-Company Login for Master/Transaction/Reports Data Access)
50,000	02. PHASE-1: Auto Email Alert Settings - 5 alert formats (Directors and IT Officers Access Only)
50,000	03. PHASE-1: Directors Dashboard
50,000	04. PHASE-1: Managers Dashboard
40,000	05. PHASE-1: Production Job Profile Master (Worksheet Data)
24,000	06. PHASE-1: Customer Master
40,000	07. PHASE-1: Machine Master with Section/Dept Creation
20,000	08. PHASE-1: Supervisors and Machine Operator Master
20,000	09. PHASE-1: Work Shift Master
15,000	10. PHASE-1: Material Type Master
15,000	11. PHASE-1: Product Sizes Grouping Master
25,000	12. PHASE-1: Mixing Section Label Printing, Transfer Note & Reports
25,000	13. PHASE-1: Extruder Section Label Printing, Transfer Note & Reports
25,000	14. PHASE-1: Printing Section Label Printing, Transfer Note & Reports
25,000	15. PHASE-1: Guesset Section Label Printing, Transfer Note & Reports
25,000	16. PHASE-1: Cutting Section Label Printing, Transfer Note & Reports
45,000	17. PHASE-1: Advanced Reports with Export to Excel (Free 3 Reports per PHASE-1 Features) (Excluding Auto Email Alerts to Directors/ Managers/ Approvers and Directors Dashboard Widgets)
604000	

4. Phase 2 Scope

25,000	18. PHASE-2: Delivery Scheduling & Reports (Raw Materials, Semi-Finished, Finished, Wastage, Sales Return Re-Issues)
20,000	19. PHASE-2: Gate Pass Printing (Raw Materials, Semi-Finished, Finished, Wastage, Sales Return Re-Issues)
35,000	20. PHASE-2: Returnable Gate Pass Printing and Tracking
25,000	21. PHASE-2: Sales Returns Note (SRN)
35,000	22. PHASE-2: Quality Check Test Report Sticker-Printing on Tested Sample (For Production Outputs and Sales Returns Investigation)
40,000	23. PHASE-2: Advanced Reports with Export to Excel
180,000	

5. Phase 3 Scope

To Be Decided	PHASE-3: QRL Supplier Quotation Request Letter (Email/Print)
	PHASE-3: PO Purchase Order (Local) (Goods and Non-Goods Services)
	PHASE-3: PO Purchase Order (TIEP)
	PHASE-3: PO Purchase Order (Import/Multi-Currency)
	PHASE-3: GRN Goods Received Note (Local PO)
	PHASE-3: GRN Goods Received Note (Imports/Multi-Currency)
	PHASE-3: BILL Supplier Bill Entry/Debit Note (Local)
	PHASE-3: BILL Supplier Bill Entry/Debit Note (TIEP/Import) ((Imports/Multi-Currency with Non-Goods Debits such as Duty/SPA/Daemorrhage, Insurance, Container, Forklift, Clearing Agent, Bank Debit Charges, Labour Chgs, etc)
	PHASE-3 PRN Purchase Return Note/Credit Note (Local GRN)
	PHASE-3 PRN Purchase Return Note/Credit Note (TIEP GRN)
	PHASE-3 PRN Purchase Return Note/Credit Note (Imports GRN)
	PHASE-3: GIN Goods Issue Note (General Stocks, Machine and Vehicle Spare Parts & Accessories, (Employees by Dept, Machines of the Section/Workshop, Customer, Supplier, Other)
	PHASE-3: GIRN Goods Issue Return Note with Returnable GatePass Number(Employees by Dept, Customer, Supplier, Other)
	PHASE-3: DGN Damaged Goods Note
	PHASE-3: DIN Damaged Goods Disposal Note (With/Without Value)
	PHASE-3: PST1 Physical Stock Take Statement by Store
	PHASE-3: PST2 Physical Stock Upload (Stocks Adjustment with 3 Approvals)
	PHASE-3: SAN: Stock Adjustment Note (with 3 Approvals)

	PHASE-3: GTN: Goods Trransfer Note (Non Production Job/General Stocks)
	PHASE-3: MRN: Material Request Note (By Production Sections)
	PHASE-3: MIN: Material Issue Note (to the Production Section)
	PHASE-3: MIRN: Material Issue Return Note (Excess/Floor/Wastage/Damaged Stocks/Semi-Finished Items from the Production Section)
	PHASE-3: STN: Factory/Production Floor various stage Stocks Trransfer Note (Sweep, Input Wastage, Input Damages, Processing Damages, Output Wastage, Output Damages, For Recycle, Other)
	PHASE-3: SPLIT Item Split/Dissemble Note (with Weighted Average Cost distribution)
	PHASE-3: MERGE Item Merge/Assembled Note (with Weighted Average Cost distribution)
	PHASE-3: GPN Gate Pass Note
	PHASE-3: RGPN Returnable Gate Pass Note
	PHASE-3: Inventory Transaction Forms Approval Work Flow (Technical Specs Certified By, Documents Checked By, Approved/Rejected/ReApproved By)
	PHASE-3: Free Basic Reports (2 Reports for each Transaction Form-Summary and Detailed)
	PHASE-3: Free Advance Reports (2 Reports for connecting more than 1 Transaction Forms-Summary and Detailed)
	PHASE-3: Optional, 360-degree Stocks Analysis Report (without Graphics)
	PHASE-3: Optional, Drilled-down Stocks Entry Movement Report (upto the Inventory original Transaction Note)
	PHASE-3: Optional, Directors/Managers/Approvers Dashboard Widget update
	PHASE-3: Mobile App with 24/7 Online Approving/Checking, Special Reports access, Chat Meetings with storekeepers
	Cloud Server Storage (To Be Decided with PHASE-1 Cloud Space)

6. Phase 4 Scope

To Be Decided	PHASE-4: Sales Module Features (To Be Listed on customer inquiry)
	PHASE-4: Sales Module - Mobile App for Salesman
	PHASE-4: Sales Module - Mobile App for Approvers/Directors

7. Phase 5 Scope

To Be Decided	PHASE-5: Financial Accounting Module Features (To Be Listed on customer inquiry)
	PHASE-5: Financial Accounting Budget Sub-module Features (To Be Listed on customer inquiry)
	PHASE-4: Financial Accounting Module - Mobile App for Accountants, Billing, Purchasing Teams
	PHASE-4: Financial Accounting Module - Mobile App for Directors

8. Phase 6 Scope

To Be Decided	PHASE-5: Assets Movement and Activity Tracking Module (Buildings, Vehicles, Office Furniture/Equipment, etc)
	PHASE-5: Assets Movement and Activity Tracking Module (Buildings, Vehicles, Office Furniture/Equipment, etc - Mobile App for Drivers, Workshop Technicians, etc
	PHASE-5: Assets Movement and Activity Tracking Module (Buildings, Vehicles, Office Furniture/Equipment, etc - Mobile App for Directors

9. Proposed Technology Stack

LAYER	TECHNOLOGY	NOTES
Frontend	React (PWA)	Mobile-accessible, installable on factory floor devices
Backend	Node.js	REST API,
Database	PostgreSQL	Cloud-hosted, no licensing cost
Hosting	Contabo VPS	Cloud Server
Security / WAF	Cloudflare (Free)	DDoS protection, HTTPS, DNS management, Reverse Proxy